[INCH-POUND]
A-A-59241
June 16, 1998
SUPERSEDING
KKK-D-2778
March 26, 1992

COMMERCIAL ITEM DESCRIPTION

DISTRIBUTORS, WATER TANK, TRUCK MOUNTED FRONT AND REAR SPRAY

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description (CID) covers water tank distributors with front and rear spray heads, rear spraybar, and mounted on a diesel-engine-driven truck. The distributors are for use in soil stabilization and soil compaction operations. The distributors can also be used for transporting or transferring water, for application of water for dust alleviation, sump pumping, and as auxiliary fire fighting equipment.

2. CLASSIFICATION. The distributors shall be of the following types as specified (see 7.2):

Type I - 1,000-gallon (gal) (3 785 litre (L)) tank capacity, 16,000 pounds (lb). (7 258 kilogram (kg)) gross vehicle weight rating (GVWR), 4 by 2.
Type II - 2,000-gal (7 571 L) tank capacity, 28,000 lb (12 701 kg) GVWR, 4 by 2.
Type III - 2,000-gal (7 571 L) tank capacity, 37,000 lb (16 783 kg) GVWR, 6 by 4.
Type IV - 2,000-gal (7 571 L) tank capacity, 46,000 lb (20 865 kg) GVWR, 6 by 6.

3. SALIENT CHARACTERISTICS.

3.1 Description. The distributor consists essentially of water tank, water pump and pump valve station, front and rear header pipes with spray heads, rear mounted spray/sprinkler bar, controls, piping, and safety devices, mounted on a diesel-engine-driven truck having a GVWR as specified herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Officer In Charge (Code 15E2), Seabee Logistics Center, 4111 San Pedro St., Port Hueneme, CA 93043-4410, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3825

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3.2 Standard commercial product. The distributors shall, as a minimum, be in accordance with the requirements of this CID and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this CID but which are a part of the manufacturer's standard commercial product, shall be included in the distributors being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Interchangeability. All distributors of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4 Design and construction.

3.4.1 Ambient temperatures. The distributor, including all components, accessories, and auxiliaries, shall operate satisfactorily at an ambient temperature range from as low as - 25 degrees Fahrenheit (°F) (- 31.6 degrees Celsius (°C)) to as high as 125 °F (51.6 °C).

3.4.2 Vehicle and components. The vehicle, including all required components and accessories, shall comply to all provisions of EPA 40 CFR 86, EPA 40 CFR 202, EPA 40 CFR 205, DoT 49 CFR 325, DoT 49 CFR 393, DoT 49 CFR 570, and DoT 49 CFR 571, as applicable, in effect on date of vehicle manufacture.

3.4.2.1 Cab. The cab shall be manufacturer's standard construction, standard full width, fully enclosed, heated, and insulated. Seats shall be upholstered and furnished with seat belts. Operator's seat shall be adjustable for height, forward, and rearward position. The window controls shall be manual operated. All glass shall be tinted safety glass. The cab shall be designed to provide the driver easy access from ground to the cab component. All exterior step surfaces shall be nonskid or grated type. The cab defrosting and heating shall be thermostat controlled with multispeed fan, capable of maintaining the cab temperature to not less than 55 °F (12.7 °C). The defroster shall keep the windshield surface clear at the specified ambient temperature.

3.4.2.2 Engine. The engine shall be manufacturer's standard diesel-fueled engine.

3.4.2.3 Transmission. The chassis power train shall include an automatic transmission with not less than four forward speeds. An auxiliary transmission or transfer case, required to meet the requirements as specified herein, shall be included.

3.4.2.4 Traction control. A traction control that actuates automatically to ensure that power is transmitted to the wheel having traction when the opposite wheel loses traction shall be provided on all rear axles. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.4.2.5 Steering. The vehicle shall have front wheel, hydraulic assisted power steering. Failure of the hydraulic system shall not prevent manual steering of the vehicle.

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3.4.2.6 Service brakes. The service brakes shall be power assisted of the vacuum-hydraulic system for type I, and a full air brake system for types II, III, and IV distributors.

3.4.2.7 Power-take-off (PTO). When furnished, the PTO shall have sufficient rating to operate the hydraulic powered equipment. A caution plate or a decal that reads “DO NOT OPERATE VEHICLE AT HIGHWAY SPEED WITH POWER-TAKE-OFF ENGAGED” shall be installed visible from the driver’s seat. A red warning light to indicate PTO engagement shall be provided at the dashboard.

3.4.2.8 Hydraulic system. The hydraulic system shall be furnished complete with all necessary safety devices, alarms, and controls for a satisfactory operation of the hydraulic driven equipment. No high pressure hydraulic flexible hoses shall be used inside the truck cab.

3.4.2.9 Operating controls. All controls required to operate the electric, pneumatic, and hydraulic components shall be properly identified, using universal symbols or English language, or both. Controls shall be located inside the cab and within reach from the driver’s position.

3.4.2.10 Electrical system. The electrical system shall operate on a 12-volt negative ground. The charging alternator shall have not less than a 75-ampere rating.

3.4.2.11 Starting system. Engine starting shall operate from a 12-volt negative ground battery(s). The battery(s) shall be standard commercial, maintenance free type with sufficient cold cranking amperes for the designed ambient temperature conditions.

3.4.2.12 Wheels and tires. Wheels shall be disc type. All wheels and tires shall be the same size and interchangeable between axles. Centerline difference between front and rear wheels shall be not greater than 1-inch (25.4 millimetre (mm)). Tires shall be wide-base tubeless steel belted radial with nondirectional mud or traction tread. Wheels and tires shall conform to TRA recommendation. When specified (see 7.2), a fully inflated spare tire and spare tire carrier shall be furnished.

3.4.2.13 Outside rear view mirrors. Outside rear view mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches (in^2) (323 square centimetre (cm^2)). The convex portion shall have not less than 20 in^2 (129 cm^2) of reflective surface.

3.4.2.14 Panel instruments. Panel instruments shall be visible from the driver’s location, and shall include not less than the following:

a. Engine hour meter (9999 reading).
b. Keyed ignition/starting switch.
c. Voltmeter or charging indicator.
d. Fuel gage.
e. Oil pressure gage.

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f. Engine coolant temperature gage.
g. Speedometer with recording odometer.
h. Low air pressure warning light and audible alarm for air brake system.

3.4.2.15 Accessories. Accessories shall include, but not limited to, the following:

a. Radiator coolant recovery system.
b. Two tow hooks at both front and rear for the purpose of towing the vehicle.
c. Fuel tank(s) capacity adequate for not less than 10 hours operation.
d. Accessories for quick engine starting in cold weather (measured shot heat sensing ether injection system with not less than 12 ounces (354 millilitre (mL)) reservoir or glow plugs).
e. Splash guards and stone throw protection.

3.4.3 Distributor components.

3.4.3.1 Water tank. The water tank shall be a welded steel construction, elliptical cylinder shape, and after fabrication, shall be hydrostatically or air tested to not less than 5 pounds per square inch gage (34.5 kilopascal (gage)). The tank shall have a top-mounted access hole with not less than 18 inches (457 mm) diameter. A sight glass type tank gage, located so as to be readable from the driver inside the cab or a mechanical type gage with accuracy of plus or minus 50 gallons (189 L) of actual amount of water inside the tank shall be provided. Tank vent and overflow connection shall be not less than 3 inches (80 mm) nominal diameter pipe.

3.4.3.2 Spray heads and spray bar. Each side of the truck (front and rear) shall be provided with remote controlled spray heads that are adjustable from not less than 8 feet (2.4 metre (m)) to not less than 24 feet (7.3 m) of spraying range. The spray heads shall be installed above the top of the vehicle bumper height. A remote controlled combination gravity/pressure type spray/sprinkler bar shall be mounted at the rear. Each spray head and the spray bar shall be capable of being operated independently or simultaneously from the driver's position.

3.4.3.3 Water pump. The water pump, powered by a PTO driven hydraulic motor, shall be self-priming, single stage, single suction centrifugal, and have a capacity of not less than 350 gallons per minute (gpm) (22 cubic metre per second (m^3/s)) when operating at 30 feet (9 m) discharge head and 25 feet (7.6 m) suction lift. Pump suction piping shall be 4-inch (102 mm) nominal diameter, complete with connections to match the discharge end of the hose specified herein.

3.4.3.4 Pump valve station. The pump valve station shall be located on the left hand side (street side) of the vehicle with the operating valves readily accessible to the operator standing on the ground. The instruction plate showing the valve manifold diagram and proper operating procedures shall be visible to the operator. All necessary piping, suction strainer, vacuum gage (suction side), pressure gage (discharge side), valves, fittings, safety device and controls, shall be complete to perform the following operations:

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a. Pump water from outside source (such as lakes, ponds, or streams) into the distributor tank.
b. Pump water from outside source to a 1.5-inch (38 mm) hose outlet, without the water entering the distributor tank.
c. Pump water from distributor tank to the spray heads.
d. Pump water from distributor tank to a 1.5-inch (38 mm) hose outlet.

3.4.3.5 Suction hose assembly. The suction hose assembly shall consist of three 10-foot (3 m) sections of 4-inch (102 mm) hose. Each section of the hose shall be fitted with a 4-inch (102 mm) quick disconnect coupling (hose end and faucet end). One double adapter shall be furnished. A 4-inch (102 mm) foot valve, complete with strainer, shall be provided for use with the suction hoses.

3.4.3.6 Discharge hose assembly. The discharge hose assembly shall be 1.5-inch (38 mm) diameter and 25 feet (7.6 m) long. A nozzle, size 1.5-inch (38 mm) with 0.625-inch (15.9 mm) tip opening shall be provided. A reducing hose connection, size 2.5-inch (64 mm) by 1.5-inch (38 mm), shall be provided to be connected to the hose end of a fire hydrant.

3.4.4 Air transportability configuration. When specified (see 7.2), the distributor shall be transportable by aircraft in a roll-on/roll-off configuration in accordance with MIL-STD-1791. The air transportability configuration shall be for air transport by C-5, C-17, C-130, or C-141 aircraft as specified (see 7.2).

3.4.5 Rail impact. When specified (see 7.2), the distributor shall meet the rail impact requirements of MIL-STD-810.

3.5 Performance.

3.5.1 Speed and gradeability. Speed requirements shall be met with the distributor loaded to specified GVWR.

3.5.1.1 High speed. When driven on a paved concrete or asphalt public road, the vehicle shall be capable of maintaining speed of not less than 55 miles per hour (mph) (89 kilometre per hour (km/h). When driven on a concrete or asphalt road with 5 percent grades, the vehicle shall ascend at speeds of not less than 45 mph (72 km/h).

3.5.1.2 Low speed. With the transmission shifted to the lowest gear and with the engine running at not less than 75 percent of recommended governed speed, the vehicle speed shall be not greater than 4.5 mph (7.2 km/h) for type I, 5.0 mph (8.0 km/h) for type II, and 3.5 mph (5.6 km/h) for types III and IV.

3.5.2 Application and distribution. The rate of application shall be not less than 120 gpm (7.6 m^3/s) when distributed through the front or rear spray heads and 465 gpm (29 m^3/s) when distributed through all spray heads and spray/sprinkler bars. There shall be an even wetting of the road surface when water is sprayed with the vehicle traveling at 3.5 mph (5.6 km/h).

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3.6 Ground clearance. The distributor, without disassembly of any component and moving at its own power, shall have ground clearance sufficient to negotiate, approach, and depart from a 100-foot (30.5 m) loading ramp in a fully lowered position and inclined at an angle of 21 degrees from the horizontal.

3.7 Safety. For personnel protection, all rotating or moving parts and parts subject to high operating temperature, shall be insulated, enclosed or guarded. Each distributor shall be furnished with backup lights and audible alarm that is activated when the transmission is shifted into reverse gear.

3.8 Air pollution control. When specified (see 7.2), trucks destined for California shall comply with the State of California's air pollution control regulation.

3.9 Overall dimension. The overall width shall be not greater than 96 inches (2 438 mm) and the overall height shall be not greater than 102 inches (2 591 mm).

3.10 Rustproofing . The vehicle shall be rustproofed and undercoated in accordance with FED-STD-297.

3.11 Tools. Each vehicle shall be furnished with tools required for exchanging mounted tire assembly with the spare assembly. Tools shall include not less than a hydraulic jack, jack handle, and wheel nut wrench. The jack shall be of such closed height to permit its location under the axle, or other satisfactory lift point at any wheel with flat tire. The jack, without blocking, shall be capable of raising any wheel of the fully loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.12 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J 534. Where use of high pressure lubricating equipment, 1,000 psi (6 895 kPa) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.13 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include not less than the following:

a. Inflation of all tires.
b. Adjustment of brakes (when required).
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of engine (when required).
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point. The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

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g. Cooling system filled to capacity with clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

3.14 Lifting and tiedown attachments. When specified (see 7.2), the truck shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the distributor. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the distributor on the carrier when shipped.

3.15 Identification marking. Identification shall be permanently and legibly marked directly on the distributor or on a corrosion-resisting metal plate securely attached to the distributor at the source of manufacture. Identification shall include the manufacturer’s model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.16 Vehicle markings. Vehicle markings shall conform to MIL-STD-1223, applicable to the branch of service or agency as specified (see 7.2).

3.17 Instruction plates. The equipment shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, bolts, or rivets.

3.18 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 7.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils (0.06 mm) over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.19 Tool box. The vehicle shall be provided with a lockable metal tool box. The tool box shall be weather tight. Tool box size shall be adequate to store all the necessary tools and fixtures as required herein.

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3.20 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of IEEE-SI-10 and all other requirements of this commercial item description including form, fit and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

7.1.2 The Federal Acquisition Regulation (FAR), Environmental Protection Agency (EPA), and Department of Transportation (DoT) regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.3 IEEE Standard is available from the Institute of Electrical and Electronics Engineers, IEEE Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

7.1.4 SAE Standards and Recommended Practices is available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

7.1.5 TRA Yearbook is available from the Tire and Rim Association, Inc., 175 Montrose Avenue, Suite 150, Copley, OH 44321.

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7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, date of this CID.
b. Type of distributor required (see 2.).
c. When a fully inflated spare tire and spare tire carrier is furnished (see 3.4.2.12).
d. When the distributor is to be configured for air transport and type of aircraft for which distributor is to be configured (see 3.4.4).
g. When the distributor is to meet the rail impact test of MIL-STD-810 (see 3.4.5).
h. When the distributor is to comply with California's air pollution control regulation (see 3.8).
i. When lifting and tiedown attachments are required (see 3.14).
j. Departmental service or agency to which vehicle markings are applicable (see 3.16).
k. Color of finish coat (see 3.18).

7.3 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN used for items acquired to this description are created as follows:

AA59241 X
CID number
Type of distributor (see 2.)

PIN codes:
1 = Type I
2 = Type II
3 = Type III
4 = Type IV

7.4 Supersession data. This commercial item description replaces Federal Specification KKK-D-2778, dated March 26, 1992.

7.5 Classification cross reference. Classifications used in this commercial item description (see 2.) are identical to those found in the superseded Federal Specification KKK-D-2778.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Review Activities:
Army - MT
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Navy - YD1

(Project 3825-0245)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59241	**2. DOCUMENT DATE (YYMMDD)** 980616

3. DOCUMENT TITLE

DISTRIBUTORS, WATER TANK, TRUCK MOUNTED FRONT AND REAR SPRAY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME R. E. PAMPLONA	b. TELEPHONE *Include Area Code)* (1) Commercial 805-982-5843 (2) AUTOVON 551-5843
c. ADDRESS *(Include Zip Code)* OFFICER IN CHARGE, SEABEE LOGISTICS CENTER (CODE 15E2), 4111 SAN PEDRO ST. PORT HUENEME, CA 93043-4410	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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